Humans as Sensors: Cost-Aware Routing for Structured Human–AI Information Gathering

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Abstract

AI agents often lack the context needed to make informed decisions. Rather than fully deferring the decision, we propose treating humans as callable sensors that agentic systems can query to reduce uncertainty. We study two modalities, a structured Human API and Implicit (conversational deferral), and evaluate them with LLM-simulated humans on MedQA cases. We find that while they have comparable accuracy (84–86.8%), they gather largely different evidence, vary in per-turn uncertainty reduction, and are complementary (each uniquely solving 5–6% of cases). Through fixed costs proportional to the share of diagnostic information, we show that structured sensing is cheaper. Finally, our experiment reveals that accuracy in structured sensing grows sublinearly with action space, motivating the design of smaller, well-curated registries.

1 Introduction

AI systems often lack the necessary context for effective, confident decision-making. This critical information may exist in the world and be accessible to humans but not to models. In a hospital, a patient can describe their symptoms, and a nurse can perform laboratory tests to inform an AI diagnostic tool. We argue that *decision deferral* can be avoided by selectively querying humans for information that reduces uncertainty, which we call *human sensing*. This perspective remodels humans as *callable functions* with associated costs and expected returns. Bhatt [2026] argues that as AI agents grow more autonomous in digital environments, they increasingly depend on humans to observe and act in the physical world—shifting the human role from "in the loop" to "on call." We formalise this intuition: the agent must select appropriate human-sensing actions from a registry to maximize information gain in a cost-aware routing problem. Recent work [Tang et al., 2026] proposes a similar "Human Tools" abstraction and evaluates it using human-centric metrics claiming it amplifies human strengths and enables strong AI leadership.

Building on the deferral framework from the human–AI collaboration literature [Madras et al., 2018, Mozannar and Sontag, 2020, Raghu et al., 2019], we utilize the routing rule $\pi(x) = \arg\min_{a\in\mathcal{A}} \; r_a(x;h) + c(a;x)$, selecting the action a that minimises the risk $r_a(x;h)$ and cost $c(a;x)$ for input x, where h denotes the human decision-maker. We expand standard deferral ($\mathcal{A} = \{a_{\text{model}}, a_{\text{human}}\}$) to include *structured human sensing actions* and conversational deferral:

$$\mathcal{A} = \{ \underbrace{a_{\text{model}}}_{\text{decide alone}} , \underbrace{a_1, \ldots, a_K}_{\text{structured sensing (Human API)}} , \underbrace{a_{\text{conv}}}_{\text{conversational deferral}} \}, \tag{1}$$

where each action a_j queries a specific human source for a specific type of information at cost c_j. The agent's objective is to minimize total sensing cost $\sum_j c_j \cdot \mathbb{1}[a_j \text{ selected}]$ while reaching sufficient diagnostic certainty. This defines a *multi-arm routing policy* in which the agent sequentially selects sensing actions based on its evolving belief state.

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Table 1: Collaboration strategies ($N = 582$ shared cases). Cost is mean sensing cost in registry points.

Condition	Acc. (%)	Cost	Turns
Baseline	71.8 ± 3.6	0.0	1.0
Baseline-Full	92.8 ± 2.0	54.2	1.0
H-API (fine)	85.1 ± 2.9	2.6	2.5
H-API (medium)	84.0 ± 2.9	7.0	2.2
H-API (coarse)	86.3 ± 2.6	13.6	2.1
Implicit (single)	86.8 ± 2.8	51.2	4.3
Implicit (multi)	83.5 ± 3.0	15.6	2.6

Table 2: Information overlap: implicit vs. H-API (fine) ($N = 479$).

Metric	Value
Impl. turns / case	3.7
Impl. actions / case	4.9
H-API actions / case	1.9
Coverage	0.478
Jaccard similarity	0.199
Novelty rate	2.7%

We have three main contributions. First, we formalise human sensing as a cost-aware, multi-action extension of binary deferral and operationalise it through a structured registry of typed sensing actions (Section 2.1). Second, we evaluate structured and conversational actions in terms of accuracy, cost, information overlap, per-turn uncertainty reduction, and complementarity (Sections 3.1–3.3). Third, we show that accuracy grows sublinearly with the size of the action space (Section 3.4). Additionally, we sketch an online adaptive version that learns from interaction via contextual bandits (Appendix F).

2 Setup and Methods

We simulate clinical diagnosis, where diagnostic context can be acquired through priced sensing actions from sources (e.g., nurses, specialists), on MedQA-USMLE [Jin et al., 2020], a 4-option MCQ medical licensing dataset. The diagnostic agent (GPT-4.1) considers age, gender, and chief complaint and uses sensing functions to reduce uncertainty for an accurate diagnosis. Humans are simulated by LLMs with access to ground-truth patient profiles. Each action j has a fixed cost c_j, derived with an LLM to be proportional to the information quantity. Implicit costs are calculated by summing the costs of all sensing actions required to obtain the context. For Implicit (multi), contexts are separated by source. Concretely, the risk is $r_a(x; h) = 1 - P(\text{correct} \mid \text{evidence from } a)$ and the cost $c(a; x) = c_j$ is the fixed registry cost.

2.1 Main Comparison

We evaluate mean accuracy, cost, and turns on 582 cases and compare human sensing (H-API) and the conversational (Implicit) against the Baseline (model alone floor) and the Baseline-Full (model with full ground-truth context ceiling). Additionally, we compare the H-API across three levels of registry granularity, with less granular actions returning more information at a higher cost. Fine is a registry of 62 atomic sensing actions. Medium has 20 actions composed of fine. The coarse includes 8 broad categories.

2.2 Information Analysis

To understand the two strategies in terms of information gathering, we perform two analyses. First, we classify the set of actions corresponding to each implicit turn with an LLM judge (GPT-4.1-mini) to compare Implicit and H-API on $N = 479$ shared cases to calculate coverage, Jaccard similarity, and novelty rate. Then, we study the per-turn information gain and uncertainty reduction by inserting diagnostic checkpoints at each sensing turn and forcing the models to make a prediction and using a separate scoring model to estimate the $P(\text{correct option} \mid \text{evidence so far})$ on a subset of 100 cases.

2.3 Complementarity Analysis

Human-AI collaboration is complementary when the team outperforms either party alone [Bansal et al., 2021]; formally, $R(\pi^\star) < \min(R_{\text{model}}, R_{\text{human}})$ under optimal deferral. We compare the agreement between H-API (fine) and Implicit (single) on 582 shared cases. For the 76 cases where they disagree, we attempt to pick the better one based on self-reported confidence and an independent LLM judge (GPT-4.1) that picks the most plausible diagnosis based on reasoning, and compare that to an oracle that always selects the better strategy.

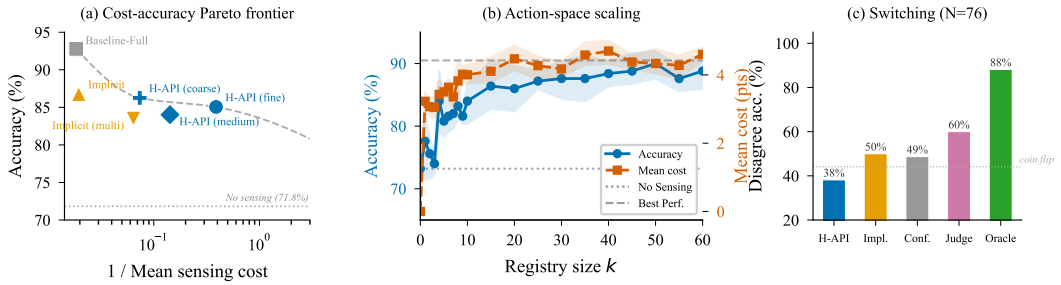

Figure 1: **(a)** Cost-accuracy Pareto frontier. **(b)** Action-space scaling. Accuracy (blue) and cost (red) vs. registry size k. **(c)** Switching signal quality on disagreements.

2.4 Action-Space Scaling

We vary the sensing registry size from $k = 0$ to $k = 60$, randomly picking k actions with replacement, in 5 repetitions of 50 cases each per registry size, to study how the action space affects accuracy and cost. Drawing $k = 60$ from 62 actions yields $\sim$39 unique actions in expectation (Appendix C).

3 Results

3.1 Main Comparison

Table 1 compares all conditions showing that H-API and Implicit perform with comparable mean accuracy (83.5% – 86.8%), which improves over the Baseline (71.8%), but does not reach Baseline-Full (92.8%). H-API spent fewer resources than Implicit at all granularities, showing the benefit of structured sensing in requesting targeted cost-aware actions. Additionally, the trend shows the total cost increases as the registry size becomes coarser. The Pareto frontier (Figure 1a) passes through H-API (coarse) and H-API (fine) between the baselines. Figure 1c shows switching signal quality on disagreement cases. All conditions show overconfidence (ECE 0.200 baseline, 0.123 H-API fine), with the risk-coverage curve nearly flat (Figure 5, Appendix), motivating active sensing over confidence-based routing.

3.2 Information Analysis

The information analysis reveals that Implicit and H-API gather largely different information (Table 2; Jaccard similarity of 0.199). Implicit utilizes 4.9 actions on average compared to 1.9 for H-API. They also differ in how quickly they reduce uncertainty. Both start at $P(\text{correct}) \approx 0.34$ (Figure 2), but H-API increases steadily to 0.57 by turn 3. H-API entropy drops from 1.77 to 1.27 bits while Implicit's plateaus near 1.4 bits after the first turn, suggesting diminishing returns from extended conversation. The small sample at later H-API checkpoints warrants caution.

3.3 Complementarity Analysis

The complementarity analysis reveals H-API and Implicit agree on 80.1% of cases, each uniquely solving 5–6% (Figure 4, Appendix), resulting in an oracle selector scoring 87.8% [85.2, 90.4]. In the cases where H-API and Implicit disagree, confidence-based switching scores 47% (slightly better than a coin flip at 44%) while the LLM judge reached 60%, improving over each independently but still far from the oracle selector (88%).

3.4 Action-Space Scaling

The action-space scaling experiment for H-API shows accuracy rises sublinearly with registry size (Figure 1b), plateauing at $87.6 \pm 3.2\%$ around $k = 30$. Cost does not grow beyond registries of 25. In all cases, the diagnostic agent utilized at most 5 actions. These results suggest that a small, well-curated registry captures most of the performance gains.

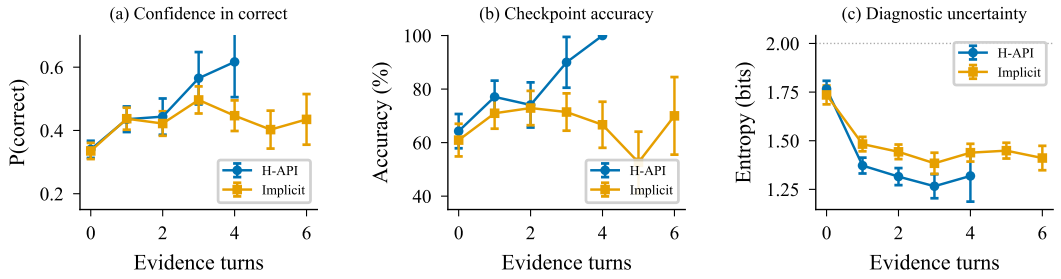

Figure 2: **(a)** $P(\text{correct})$ vs. evidence turns. H-API shows steady gains; implicit stalls after turn 1. **(b)** Forced-diagnosis accuracy per turn. **(c)** Diagnostic entropy.

4 Discussion

From deferral to sensing. Unlike binary deferral [Mozannar and Sontag, 2020, Madras et al., 2018], the agent dynamically selects *which* human sensor to query based on its current belief state. We hypothesize that structured human sensing may redefine the human role in existing workflows, yielding benefits such as reduced workload or algorithmic upskilling, but also risks cognitive offloading [Gerlich, 2025], skill atrophy [Macnamara et al., 2024], and meaning erosion. Future work should focus on developing human-centric designs and validating them through real-world human experiments.

When sensing hurts. In approximately 5% of cases, sensing decreases accuracy due to noisy or misleading responses. The optimal policy must account for when human input degrades decision-making. Identifying when to abstain from sensing is an open problem for future work.

Online adaptation. In deployment, the fixed routing policy could be improved via online adaptation that learns from interaction. A contextual bandit [Li et al., 2010] can learn which sensing actions to select from per-case diagnostic feedback while accumulating aggregate statistics about actions and sources that enrich the LLM's context over time. Exploration is budget-constrained: at each turn the agent restricts its choices to the set of affordable actions given the remaining budget, and an ϵ-greedy rule biased toward under-explored actions balances exploration with the LLM's own exploitation policy. We formalise this adaptive loop in Appendix F.

Limitations. Our study has several strong limitations. Human sensors are simulated with LLMs that may not accurately represent the cognitive, functional, and physical capabilities of real people. Furthermore, we evaluate a single domain and model provider. The action registries are generated by LLMs, and the costs used are fixed points rather than real-world prices. All experiments are sensitive to the prompts (released in Appendix E for reproducibility), and verbalized confidence is known to be poorly calibrated [Xiong et al., 2024, Groot and Valdenegro Toro, 2024]. The sample sizes are small (582 cases, 100-case information subset), limiting statistical strength. Finally, the bandit algorithm (Appendix F) has not been validated empirically and assumes stationarity that may not hold in practice [Li et al., 2010].

Conclusion. Motivated by a transition from AI-assisted to human-assisted workflows, we proposed exposing humans as callable sensors via a structured API and studied this approach in terms of accuracy, cost, information gathering, and uncertainty reduction. The simulated experiment showed increased accuracy over the baseline and reduced cost compared to the free-form conversational interface. Additionally, we explored how the action space affects performance and found that smaller, well-curated sensing registries capture most of the accuracy gains cost-effectively. Finally, we discussed an online, adaptive version that learns action and source utility through interaction. We hope our framework motivates real-world implementations and future research on it, particularly regarding its human-centric effects.

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A Framework Overview

Figure 3 illustrates the humans-as-sensors framework. The AI agent receives a patient vignette and makes a diagnosis by querying the humans either through structured sensing actions or conversational deferral.

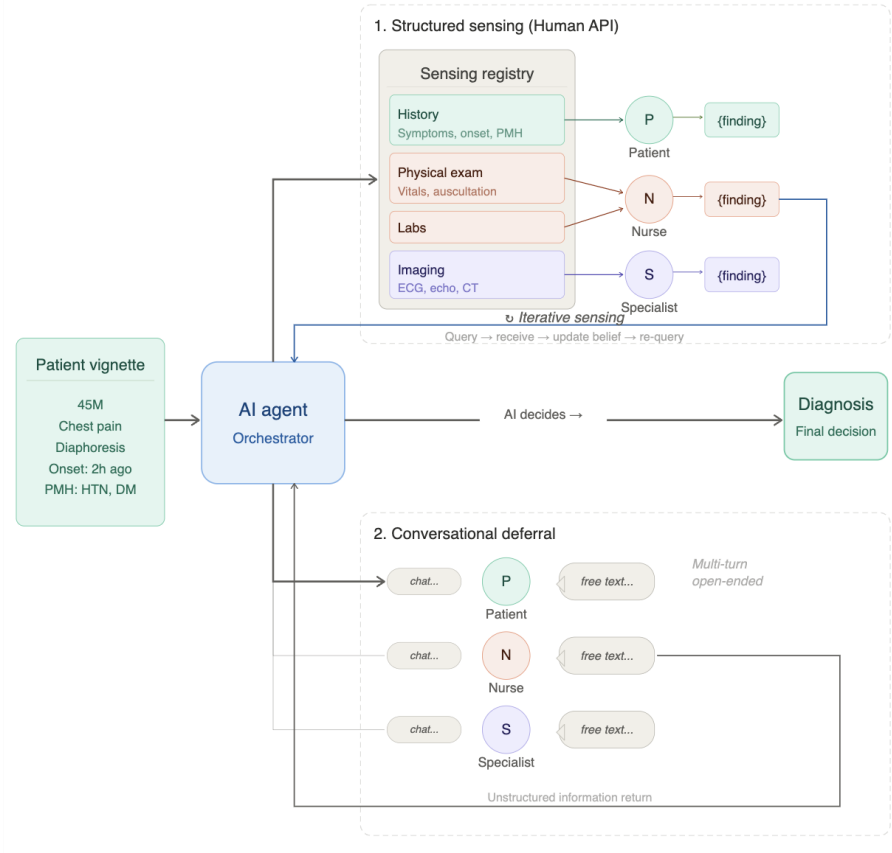

Figure 3: The humans-as-sensors framework.

B Additional Experimental Results

Case-level agreement. Figure 4 shows the case-level agreement between H-API (fine) and Implicit (single). Each strategy uniquely solves 5–6% of cases.

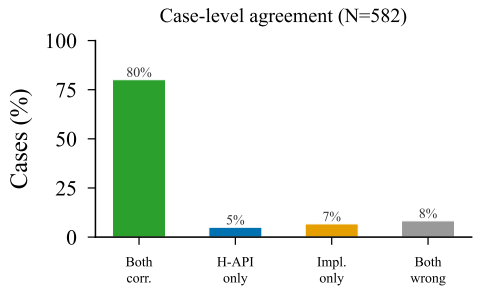

Figure 4: Case-level agreement between H-API and Implicit.

Calibration and risk-coverage. Figure 5 shows risk-coverage curves for the baseline and H-API (fine).

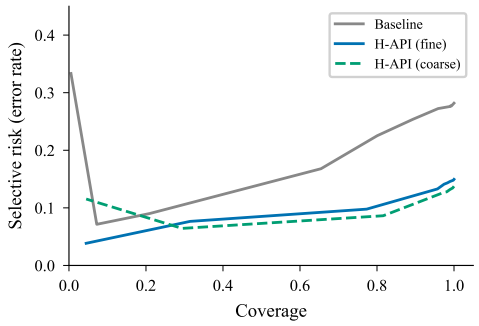

Figure 5: Risk-coverage curves.

Cost distribution. Figure 6 shows the distribution of total sensing cost across granularity levels.

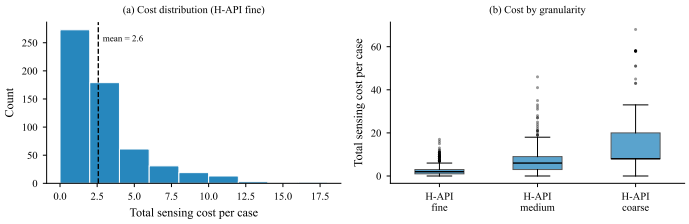

Figure 6: Cost distribution across sensing granularities.

Action co-occurrence. Figure 7 shows which sensing actions are frequently selected together.

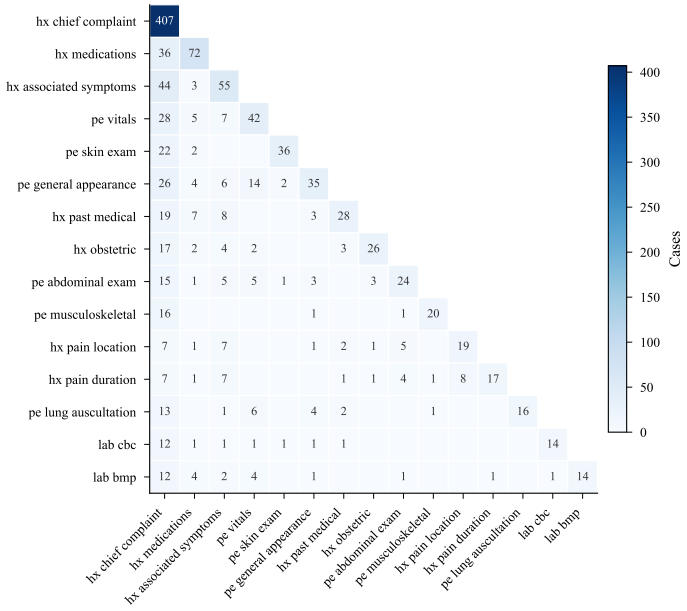

Figure 7: Co-occurrence matrix of sensing actions selected by H-API (fine).

C Expected Unique Actions under Sampling with Replacement

When drawing k actions uniformly with replacement from a registry of n actions, the expected number of unique actions is

$$\mathbb{E}[\text{unique}] = n\left(1 - \left(1 - \tfrac{1}{n}\right)^k\right) = 62\left(1 - \left(\tfrac{61}{62}\right)^{60}\right) = 62 \times 0.6230 \approx 38.6. \tag{2}$$

D Sensing Action Registries

We construct sensing action registries at three granularity levels. Each action has a category, cost (in points), and source (which human role provides the information). Table 3 shows the fine-grained registry (62 actions). Tables 4 and 5 show medium (20 actions) and coarse (8 actions) registries formed by bundling fine-grained actions.

Table 3: Fine-grained sensing action registry (62 actions). ID, name, category, cost (points), and source.

ID	Name	Cat.	Src	Cost
History Taking (20 actions, cost 1–3)				
hx_chief_complaint	Chief Complaint Clarification	Hx	patient	1
hx_pain_location	Pain Location	Hx	patient	1
hx_pain_character	Pain Character	Hx	patient	1
hx_pain_duration	Pain Duration & Onset	Hx	patient	1
hx_aggravating_relieving	Aggravating/Relieving Factors	Hx	patient	1
hx_associated_symptoms	Associated Symptoms	Hx	patient	2
hx_medications	Current Medications	Hx	patient	1
hx_allergies	Allergies	Hx	patient	1
hx_past_medical	Past Medical History	Hx	patient	2
hx_family	Family History	Hx	patient	2
hx_social	Social History	Hx	patient	2
hx_review_of_systems	Review of Systems	Hx	patient	3
hx_travel	Travel History	Hx	patient	1
hx_sexual	Sexual History	Hx	patient	2
hx_obstetric	Obstetric History	Hx	patient	2
hx_dietary	Dietary History	Hx	patient	1
hx_immunization	Immunization History	Hx	patient	1
hx_psychiatric	Psychiatric History	Hx	patient	2
hx_substance_detail	Substance Use Detail	Hx	patient	2
hx_exposure	Exposure History	Hx	patient	1
Physical Examination (17 actions, cost 2–3)				
pe_vitals	Vital Signs	PE	nurse	2
pe_general_appearance	General Appearance	PE	nurse	2
pe_cardiac_auscultation	Cardiac Auscultation	PE	nurse	2
pe_lung_auscultation	Lung Auscultation	PE	nurse	2
pe_abdominal_exam	Abdominal Examination	PE	nurse	2
pe_neurological	Neurological Exam	PE	specialist	3
pe_skin_exam	Skin Examination	PE	nurse	2
pe_musculoskeletal	Musculoskeletal Exam	PE	nurse	2
pe_lymph_nodes	Lymph Node Examination	PE	nurse	2
pe_ent	ENT Examination	PE	nurse	2
pe_eye	Eye Examination	PE	nurse	2
pe_breast	Breast Examination	PE	nurse	2
pe_rectal	Rectal Examination	PE	nurse	2
pe_peripheral_pulses	Peripheral Pulses	PE	nurse	2
pe_thyroid	Thyroid Examination	PE	nurse	2
pe_mental_status	Mental Status Exam	PE	specialist	3
pe_gait_balance	Gait & Balance Assessment	PE	nurse	2
Laboratory Tests (14 actions, cost 4–8)				
lab_cbc	Complete Blood Count	Lab	nurse	4
lab_bmp	Basic Metabolic Panel	Lab	nurse	4
lab_liver_panel	Liver Function Panel	Lab	nurse	5
lab_troponin	Troponin	Lab	nurse	5
lab_d_dimer	D-dimer	Lab	nurse	5
lab_urinalysis	Urinalysis	Lab	nurse	4
lab_tsh	TSH	Lab	nurse	4
lab_lipid_panel	Lipid Panel	Lab	nurse	4

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Table 3 continued from previous page

ID	Name	Cat.	Src	Cost
lab_coagulation	Coagulation Studies	Lab	nurse	5
lab_blood_culture	Blood Culture	Lab	nurse	6
lab_hba1c	Hemoglobin A1c	Lab	nurse	4
lab_bnp	BNP / NT-proBNP	Lab	nurse	5
lab_abg	Arterial Blood Gas	Lab	nurse	6
lab_csf	CSF Analysis	Lab	nurse	8
Imaging Studies (11 actions, cost 6–15)				
img_chest_xray	Chest X-ray	Img	specialist	6
img_ecg	Electrocardiogram	Img	specialist	6
img_abdominal_us	Abdominal Ultrasound	Img	specialist	8
img_ct_angiogram	CT Angiogram	Img	specialist	15
img_ct_head	CT Head	Img	specialist	10
img_mri_brain	MRI Brain	Img	specialist	15
img_echocardiogram	Echocardiogram	Img	specialist	10
img_extremity_xray	Extremity X-ray	Img	specialist	6
img_ct_abdomen	CT Abdomen/Pelvis	Img	specialist	12
img_doppler_us	Doppler Ultrasound	Img	specialist	8
img_mammogram	Mammogram	Img	specialist	8

Table 4: Medium-grained registry (20 bundled actions). Each action bundles 2–8 fine-grained actions.

ID	Name	Cost	Cat.	Src
History (5)				
hx_presenting_illness	History of Present Illness	6	Hx	patient
hx_medications_allergies	Medications & Allergies	2	Hx	patient
hx_medical_background	Medical, Family & Social History	5	Hx	patient
hx_review_of_systems	Review of Systems	3	Hx	patient
hx_specialized_screening	Specialized History Screening	9	Hx	patient
Physical Exam (5)				
pe_vitals_general	Vital Signs & General Assessment	3	PE	nurse
pe_cardiopulmonary	Cardiopulmonary Exam	4	PE	nurse
pe_abdominal	Abdominal & GI Exam	4	PE	nurse
pe_neuropsychiatric	Neurological & Mental Status Exam	7	PE	specialist
pe_head_neck_peripheral	Head, Neck & Peripheral Systems Exam	12	PE	nurse
Labs (5)				
lab_basic_panel	Basic Laboratory Panel	10	Lab	nurse
lab_metabolic_endocrine	Metabolic & Endocrine Panel	14	Lab	nurse
lab_cardiac_coag	Cardiac & Coagulation Markers	16	Lab	nurse
lab_cultures	Blood Culture	6	Lab	nurse
lab_invasive	Arterial Blood Gas & CSF Analysis	12	Lab	nurse
Imaging (5)				
img_ecg	Electrocardiogram	6	Img	specialist
img_plain_films	Plain Film Radiographs	10	Img	specialist
img_ultrasound	Ultrasound Studies	18	Img	specialist
img_ct	CT Studies	28	Img	specialist
img_mri_echo	MRI & Echocardiography	20	Img	specialist

Table 5: Coarse-grained registry (8 domain-level actions). Each action bundles an entire clinical domain.

ID	Name	Cost	Cat.	Src
hx_focused	Focused History (HPI + Medications)	8	Hx	patient
hx_comprehensive	Comprehensive Background History	18	Hx	patient
pe_complete	Complete Physical Examination	25	PE	nurse
lab_standard	Standard Laboratory Workup	25	Lab	nurse
lab_acute	Acute & Specialized Lab Workup	25	Lab	nurse
img_basic	Basic Imaging & ECG	12	Img	specialist
img_ultrasound	Comprehensive Ultrasound & Echo	25	Img	specialist
img_advanced	Advanced Cross-Sectional Imaging	35	Img	specialist

E Prompt Templates

E.1 Baseline Agent

```
You are an expert diagnostic physician. You will be given a
patient's triage information (age, gender, chief complaint)
and a multiple-choice diagnostic question. Analyze the case
carefully and provide your diagnosis.

You must select one of the provided answer options.
```

E.2 H-API Agent

```
You are an expert diagnostic physician using a structured
clinical sensing API. You have ONLY the patient's triage
information (age, gender, chief complaint). You MUST use
sensing actions to gather the clinical details needed for
diagnosis.

On each turn you must choose ONE of two actions:
1. SENSE: Select a sensing action from the registry.
2. DIAGNOSE: Provide your final diagnosis and select an
   answer option.

Each sensing action has an associated cost. You should
minimize total sensing cost while gathering enough
information for a correct diagnosis. Prioritize high-value,
low-cost actions and avoid redundant queries.
```

E.3 Implicit Agent (Multi-Source)

```
You are an expert diagnostic physician. You have ONLY the
patient's triage information. You MUST ask questions to
gather clinical details needed for diagnosis.

On each turn you must choose ONE of two actions:
1. ASK: Ask a free-form question to a specific source.
2. DIAGNOSE: Provide your final diagnosis.

Available Sources & Context Costs:
  - patient (cost {patient_cost}): symptoms, history,
    medications, allergies, family/social history.
  - nurse (cost {nurse_cost}): physical exam findings,
    vitals, lab results.
  - specialist (cost {specialist_cost}): neurological exam,
    mental status, all imaging results.

Each source has a one-time context cost (charged on first
use). Follow-up questions to the same source are free.
```

E.4 Simulated Human Sources

E.5 Structured Output Schemas

```
class DiagnosisOutput(BaseModel):
    primary_diagnosis: str      # top-1 diagnosis
    confidence: float           # 0.0--1.0
    differential: list[str]     # up to 3 alternatives
    reasoning: str              # brief reasoning chain
    selected_option: str | None # MCQ option letter
```

F Online Adaptive Sensing via Contextual Bandits

In deployment, the AI agent can be informed by past interaction in the shape of aggregate statistics S_t accumulated across cases and injected into its context.

Safe exploration. ϵ-greedy exploration [Sutton and Barto, 2018] over the budget-feasible action set:

$$\mathcal{A}_{\text{afford}} = \{a \in \mathcal{A} : \text{cost}_t + c(a) \leq B\}. \tag{3}$$

With probability ϵ_0, the system explores by sampling from $\mathcal{A}_{\text{afford}}$ with $P(a) \propto 1/(n_a + 1)$, naturally favouring under-explored actions. Thompson sampling [Agrawal and Goyal, 2013] or LinUCB [Li et al., 2010] could replace ϵ-greedy.

Algorithm. Algorithm 1 formalises the loop.

Algorithm 1 LLM-driven adaptive sensing with aggregate statistics

Require: LLM diagnostic agent, registry $\mathcal{A}$, budget B, exploration rate ϵ_0
Initialise $S_0 \leftarrow \emptyset$ ▷ Empty statistics table
for each new case x_t, $t = 1, 2, \ldots$ **do**
  LLM receives triage(x_t) + registry $\mathcal{A}$ + summary of S_{t-1}
  $\text{cost}_t \leftarrow 0$; $\mathcal{H}_t \leftarrow \emptyset$
  while LLM chooses SENSE **and** $\text{cost}_t < B$ **do**
    $\mathcal{A}_{\text{afford}} \leftarrow \{a \in \mathcal{A} : \text{cost}_t + c(a) \leq B\}$ ▷ Eq. 3
    if $U \sim \text{Uniform}(0, 1) < \epsilon_0$ **then**
      $a_t \sim \mathcal{A}_{\text{afford}}$ with $P(a) \propto 1/(n_a + 1)$ ▷ Explore
    else
      LLM selects $a_t \in \mathcal{A}_{\text{afford}}$ ▷ Exploit
    end if
    Execute a_t; LLM receives response; $\text{cost}_t \leftarrow \text{cost}_t + c(a_t)$
    $\mathcal{H}_t \leftarrow \mathcal{H}_t \cup \{a_t\}$
    LLM decides SENSE again or DIAGNOSE
  end while
  LLM diagnoses $\hat{y}_t$; observe outcome y_t
  Update S_t: for each $a \in \mathcal{H}_t$, update $n_a \leftarrow n_a + 1$
end for